

## Project Overview

This project explores approximately 102,000 NYC Airbnb listings to understand pricing patterns, room type distribution, borough-level trends, and review activity over time. The goal was to clean a raw, messy real-world dataset and extract insights that a host, guest, or market analyst could act on.

**Guiding questions:** How are prices distributed, and does room type actually predict price? Which boroughs dominate the market? What do review patterns over time reveal, including possible data quality issues? Does host identity verification relate to price or reviews?

101,400

LISTINGS ANALYZED

\$625

MEDIAN PRICE / NIGHT

5

BOROUGHS COVERED

4

ROOM TYPES

## Dataset

### Rows

≈ 102,599 raw → ≈ 101,400 after cleaning

### Columns

26

### Key fields

price, room type, borough, minimum nights, number of reviews, last review date, availability

### Source

NYC Airbnb listings dataset

## Methodology

- 1 Standardized column names and converted last\_review to datetime.
- 2 Filled missing reviews\_per\_month with 0; dropped rows missing a listing name or host name.
- 3 Removed columns that were mostly empty (house\_rules, license) and dropped duplicate rows.
- 4 Converted price and service\_fee from currency strings to numeric values.
- 5 Ran summary statistics (df.describe()) to check the cleaned data for sanity.

## Summary Statistics (Post-Cleaning)

| Metric    | Price (\$) | Min. Nights | Reviews | Availability (days/yr) |
|-----------|------------|-------------|---------|------------------------|
| Mean      | 625.4      | 8.1         | 27.5    | 141.2                  |
| Median    | 625.0      | 3.0         | 7.0     | 96.0                   |
| Min       | 50.0       | -1,223*     | 0.0     | -10*                   |
| Max       | 1,200.0    | 5,645       | 1,024   | 3,677*                 |
| Std. Dev. | 331.6      | 30.4        | 49.5    | 135.4                  |

\*Flagged as out-of-range / impossible values — see Data Quality Notes on page 2.

## Key Insights

### PRICE SPREAD

Broadly spread across \$50–\$1,200 with no single dominant “typical” price — the market is highly heterogeneous.

### ROOM TYPE MIX

Entire home/apt is the most listed type, closely followed by private room. Shared and hotel rooms are rare.

### LOCATION

Manhattan and Brooklyn dominate listing volume; Queens, the Bronx, and Staten Island have far fewer listings.

### PRICE VS. ROOM TYPE

Median prices are surprisingly close (~\$620–\$655) across all room types — shared rooms are not meaningfully cheaper than entire homes, which is unusual and worth further investigation.

### REVIEWS OVER TIME

Two unusual spikes stand out (~2019 and ~2022) that don't match typical steady review growth, likely tied to missing or defaulted review dates rather than real booking activity.

## Data Quality Notes

### ■ Flagged, Not Modified

Reviewing the post-cleaning summary statistics surfaced three out-of-range values that were noted but left unmodified in this pass, so the dataset reflects the source as-is:

- `minimum_nights` has a minimum of -1,223, which is not physically possible.
- `last_review` includes a date of 2058-06, which is 32 years in the future.
- `availability_365` ranges from -10 to 3,677, outside its expected 0–365 bound.

These are flagged here for transparency and would be corrected (via clipping or removal) in a production analysis.

## Limitations

- The near-identical median prices across room types suggest the price column may not be a clean like-for-like price; this would benefit from validation against the original source.
- Review-date spikes suggest some `last_review` values may be placeholders rather than real dates.

## Next Steps

- Neighbourhood-level (not just borough-level) price mapping.
- A simple price prediction model using room type, borough, and availability as features.
- Geographic visualization using latitude/longitude.
- Correct the flagged out-of-range values and re-run summary statistics.

**Tools:** Python · Pandas · NumPy · Matplotlib · Seaborn | **Full analysis and charts:** `nyc_airbnb_eda.ipynb`